

KENNEDY MWANZIA

Machine Learning & Software Engineer · Retrieval, Ranking, Evaluation & Distributed Systems

Nairobi, Kenya · +254 796 516 595 · kennedymwanzia001@gmail.com

PROFESSIONAL SUMMARY

Machine learning and software engineer. I build the parts of an ML system that are usually bought: an HNSW vector index written from scratch, a two-tower retrieval model trained with sampled softmax, a point-in-time feature store, a calibrated gradient-boosted ranker, Thompson sampling for exploration, a sequential test that stops early without inflating alpha, and drift monitoring that gates a rollout. I serve them behind FastAPI at a measured p95 of 10.7 ms, put React and Next.js in front of them, and back everything with deterministic tests: **1,550+ across five platforms**, 204 of them under *mypy --strict* on the ranking system alone. I also author adversarially hardened benchmark tasks measuring whether frontier models can do real engineering work, which is the same discipline pointed at somebody else's system.

EXPERIENCE

Software Engineer & Software Instructor

Aug 2025 – Present

Techsavanna Company · Nairobi

- **Take ML and AI features to production.** Retrieval and ranking behind stable service boundaries, model outputs validated rather than trusted, calibration so that a score means a probability, explicit confidence handling and human review as the fallback path.
- **Own the model lifecycle, not only the model.** Offline evaluation against held-out data, staged promotion through shadow and canary to live, drift monitoring on live features, and a rollback that is one call rather than a redeployment.
- **Deliver full stack, from interface to data store.** React and Next.js front ends against Java, Node and Python services, covering component and state design, API contracts, relational modelling and the integration between the layers rather than one side of it.
- **Build multi-tenant SaaS on an architecture that scales:** tenant isolation and authorisation enforced at the framework layer, service decomposition with database-per-service ownership, asynchronous messaging between contexts, idempotent write paths and defined retry and failure behaviour instead of best-effort calls.
- **Lead architecture and code review** with written technical rationale covering correctness, production cost and the recommended alternative, and deliver full-stack engineering training across Java/Spring Boot, React, Node.js and Laravel.

AI Benchmark & Verifier Engineer (Expert)

2026 – Present

AfterQuery · Frontier Bench / Kepler Project

- Author **self-contained benchmark bundles** containing instruction, Dockerised environment, reference solution, sealed verifier and metadata, evaluating frontier AI agents on realistic software, ML and systems engineering work. Two bundles accepted and paid.
- **Build deterministic, tamper-resistant graders.** Root-authoritative grading in a process the candidate's code never runs in, sealed reference data unreadable to the agent, and worlds redrawn from a system-random seed per run so no answer can be precomputed.
- **Validate by adversarial arms, not assertion.** For every task I build the arms that should fail and the arms that should pass, including independently written correct solutions unlike the reference. Recent bundle: 11 arms, oracle 1.000, every bypass 0.000.

Full-Stack Software Developer

May – Aug 2025

Techsavanna Software · Nairobi

- Built a multi-tenant membership and content platform end to end, TypeScript and Express over Sequelize and MySQL with a React front end, roughly 80 backend modules, tenancy enforced at the repository layer, and per-module Zod validation with JWT authentication.

SELECTED ENGINEERING PROJECTS

Sifa: Retrieval, Ranking and Experimentation Platform

2026 – Present

Python 3.12 · NumPy · SciPy · scikit-learn · FastAPI · Pydantic · pytest · mypy (strict) · Next.js · Tailwind CSS · Docker

A personalised feed built end to end without a vector database, a feature platform or a managed experiment service. The point of the project is the parts that are usually bought: the approximate nearest neighbour index, the point-in-time feature store, the calibrated ranker, the sequential test and the rollout guard are each implemented here and measured.

- **HNSW vector index written from scratch:** layered graph, greedy descent and the neighbour selection heuristic. Recall at 10 measured against exhaustive search over the same corpus, 0.820 at ef_search 32 rising to 1.000 at 256.
- **Two-tower retrieval trained with sampled softmax** over in-batch negatives, loss falling from 2.199 to 0.336 across 12 epochs, feeding a **gradient-boosted ranker at 0.973 held-out AUC with Platt scaling** fitted on a held-out slice, so a score is a probability rather than an ordering.
- **Point-in-time correct feature store** that refuses to read a value recorded after the request it is serving, which is the difference between an offline number that survives production and one that does not.
- **Exploration and experimentation:** Thompson sampling over topics with decay, hash-bucketed assignment, and a **mixture SPRT that stops early without inflating alpha**, asserted over 300 simulated A/A runs with repeated peeking.
- **Monitoring and staged rollout:** PSI and KS drift on live features, a rollout guard on click-through, calibration and latency, and a registry enforcing draft to shadow to canary to live with one-call rollback. Served at **p50 7.5 ms, p95 10.7 ms, p99 12.6 ms** over 400 requests, on 204 tests under mypy strict.
- **Reports where it loses.** At 16,000 vectors the pure-Python graph is slower than a single BLAS matmul; the crossover is measured above 50,000 vectors and stated rather than hidden behind a recall figure.

Aegis: Self-Governing HR Automation Platform

2026 – Present

Python 3.12 · FastAPI · scikit-learn · NumPy · SciPy · Pydantic · pytest · mypy (strict) · ruff · Docker

- **Supervised attrition modelling in scikit-learn.** Gradient boosting, random forest and logistic regression behind a standardised pipeline, feature engineering that never sees absolute salary, ranked driver factors on every score, and training that raises rather than fits on under forty rows or a single outcome class.
- **Model verification and drift detection.** Determinism probing across repeated runs, PSI and a two-sample Kolmogorov–Smirnov test, combined into a PASS/WARN/BLOCK deployment gate weighting self-contradiction above input movement. It verifies an LLM, a credit model or a pricing engine unmodified.
- **Adverse-impact testing** by the four-fifths rule with a chi-squared contingency test, returning INSUFFICIENT_DATA rather than a pass when a group is too small to be evidence. **352 tests under mypy strict**, on a hash-chained audit ledger.

SmartSeason: Agricultural Operations Platform

May – Aug 2026

Java 21 · Spring Boot 3.5 · Spring Cloud Gateway · PostgreSQL · Apache Kafka · Redis · Next.js 16 · Docker

- **27 bounded contexts across 1,591 Java sources**, database-per-service behind a gateway doing JWT pre-verification, circuit breaking and Redis rate limiting. **630 tests across 28 Maven build units.**
- **Event-driven fraud detection.** Five detectors consume attendance events over Kafka and open evidence-backed cases with payout holds; confidences compound probabilistically, so two independent 0.5 signals give 0.75 rather than 1.0.
- **Money as a financial system:** double-entry ledger with idempotent replay, M-Pesa STK push and callbacks, an order saga whose compensation unwinds in reverse, and a transactional outbox closing the crash window between commit and publish.

SmartRE Kenya: Real-Estate Trust Platform

Jun – Aug 2026

Java 21 · Spring Boot 3 · PostgreSQL · PgBouncer · Apache Kafka · Kubernetes · Next.js 14 · Playwright · Docker

- **Nine Spring Boot microservices and 343 tests**, each with its own PostgreSQL database behind PgBouncer transaction pooling, and a Next.js front end across 30 routes with Playwright end-to-end specs.
- **Payments as a reliability-critical subsystem:** M-Pesa escrow with idempotency keys, distributed locking, reconciliation on late callbacks and an outbox sweeper, alongside AI-assisted document intelligence with human-review fallback.

Forecourt: Revenue Assurance for Car Washes

Jul – Sep 2026

TypeScript · Express · PostgreSQL · Zod · vitest · supertest · M-Pesa Daraja · Docker

- **Three independent ledgers reconciled** for demand, work and money, each witnessed by at least two sources and at least one that is not a human, so the product is the discrepancy between records rather than the records.
- Rules-driven reconciliation over a typed domain of jobs, plates and money, with Daraja ingestion and payment matching, covered by 38 test cases under vitest and supertest.

TECHNICAL SKILLS

Machine Learning two-tower retrieval · sampled softmax over in-batch negatives · HNSW and approximate nearest neighbour search · gradient-boosted ranking · Platt scaling and calibration · point-in-time feature stores · Thompson sampling · MMR diversification · offline and online evaluation

Statistics & Experimentation mixture SPRT and sequential testing · A/A validation · hash-bucketed assignment · PSI and Kolmogorov–Smirnov drift detection · chi-squared testing · four-fifths-rule adverse impact · determinism probing

ML Engineering scikit-learn · NumPy · SciPy · FastAPI model serving · model registry with shadow, canary and rollback · drift and calibration gating · feature engineering · threshold calibration

Languages Python · Java 21 · TypeScript · JavaScript (ES6+) · SQL · PHP

Full-Stack Next.js 16 · React 19 · Tailwind CSS · TanStack Query · Zustand · Node.js · Express.js · Spring Boot 3 · Laravel

Data & Messaging PostgreSQL · MySQL · Redis · PgBouncer · Flyway · Apache Kafka · RabbitMQ

Architecture microservices · database-per-service · event-driven design · transactional outbox · saga with compensation · multi-tenancy · RFC 7807

Testing & DevOps pytest · JUnit · vitest · Playwright · mypy (strict) · ruff · Docker · Kubernetes · CI/CD · AWS · Prometheus · Grafana · Loki

EDUCATION

BSc Information Technology · KCA University, Nairobi

2021 – 2025

Software engineering, systems design, database administration and cloud computing. Microservices Architecture, a professional course with hands-on practice, 2024. AWS Certified Cloud Practitioner, in progress.